COMPUTING FUNDAMENTALS: HARDWARE, SOFTWARE, NETWORKS AND CLOUD

DATA-OPERATION DECISION BOARD → COMPLETE COMPUTING STACK → CPU AND MEMORY HIERARCHY → PROCESSOR AND EMBEDDED MAP → PROGRAM AND EXECUTION LADDER → NETWORK DEVICE AND LAYER

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

◆ PRIMARY CORE FLOW ◈ SUBORDINATE EXTRA ★ APPROVAL / REVIEW

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

Approval: FALSE • Pending user review • source generation g2 and all prior artifacts unchanged

00

STAGE 00

DATA-OPERATION DECISION BOARD

DATA-OPERATION DECISION BOARD · FIXED DIGEST · CONFIDENTIALITY WITH KEY · BIT / BYTE → QUANTITY

CONCEPT / ARCHITECTURE
- BIT / BYTE → quantity
- ENCODING → representation

MECHANISM / APPLICATION
- COMPRESSION → size
- HASHING → fixed digest

READINESS / STATUS LIMIT
- ENCRYPTION → confidentiality with key
- BIT / BYTE → quantity

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: HASHING → fixed digest; ENCRYPTION → confidentiality with key.

01

STAGE 01

COMPLETE COMPUTING STACK

COMPLETE COMPUTING STACK · SEMICONDUCTOR + CPU + MEMORY + STORAGE + I/O · FIRMWARE + OS + DRIVER + FILE SYSTEM · PROGRAM + DATABASE + SERVICE · PROTOCOLS + DEVICES + INFRASTRUCTURE · SECURITY + ACCESS + ENERGY + E-WASTE

PHYSICAL → semiconductor + CPU + memory + storage + I/O ⇒ SYSTEM → firmware + OS + driver + file system ⇒ APPLICATION → program + database + service ⇒ NETWORK → protocols + devices + infrastructure ⇒ GOVERNANCE → security + access + energy + e-waste

CONCEPT / ARCHITECTURE
- PHYSICAL → semiconductor + CPU + memory + storage + I/O
- SYSTEM → firmware + OS + driver + file system

MECHANISM / APPLICATION
- APPLICATION → program + database + service
- NETWORK → protocols + devices + infrastructure

READINESS / STATUS LIMIT
- GOVERNANCE → security + access + energy + e-waste
- PHYSICAL → semiconductor + CPU + memory + storage + I/O

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: GOVERNANCE → security + access + energy + e-waste.

02

STAGE 02

CPU AND MEMORY HIERARCHY

CPU AND MEMORY HIERARCHY · SMALL ---------- SLOW · WORKLOAD + IPC + BANDWIDTH + THERMALS · CLOCK RATE ALONE

FETCH → DECODE → EXECUTE → STORE

REGISTER → CACHE → RAM → SSD/HDD ⇒ FAST / SMALL ---------- SLOW / LARGE ⇒ PERFORMANCE → workload + IPC + bandwidth + thermals

CONCEPT / ARCHITECTURE
- FETCH → DECODE → EXECUTE → STORE
- REGISTER → CACHE → RAM → SSD/HDD

MECHANISM / APPLICATION
- FAST / SMALL ---------- SLOW / LARGE
- PERFORMANCE → workload + IPC + bandwidth + thermals

READINESS / STATUS LIMIT
- CLOCK RATE ALONE → insufficient
- FETCH → DECODE → EXECUTE → STORE

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PERFORMANCE → workload + IPC + bandwidth + thermals.

03

STAGE 03

PROCESSOR AND EMBEDDED MAP

PROCESSOR AND EMBEDDED MAP · GENERAL CPU · CPU + MEMORY + INTERFACES · MULTIPLE SYSTEM COMPONENTS · PARALLEL WORKLOAD

MICROPROCESSOR → general CPU

MICROCONTROLLER → CPU + memory + interfaces ⇒ SoC → multiple system components ⇒ GPU / ACCELERATOR → parallel workload

STARTING CONDITION / INPUT
- MICROPROCESSOR → general CPU
- MICROCONTROLLER → CPU + memory + interfaces
- SoC → multiple system components

SCIENTIFIC OR ENGINEERING MECHANISM
- GPU / ACCELERATOR → parallel workload
- SENSOR → measure
- PROCESSOR → compute

OUTPUT / FAILURE BOUNDARY
- ACTUATOR → act

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SENSOR → measure | PROCESSOR → compute | ACTUATOR → act.

04

STAGE 04

PROGRAM AND EXECUTION LADDER

PROGRAM AND EXECUTION LADDER · OVERLAPPING PROGRESS · SIMULTANEOUS EXECUTION · COORDINATION FAILURES

SOURCE → compiler / interpreter / assembler

PROGRAM → PROCESS → THREAD ⇒ CONCURRENCY → overlapping progress ⇒ PARALLELISM → simultaneous execution

STARTING CONDITION / INPUT
- SOURCE → compiler / interpreter / assembler
- PROGRAM → PROCESS → THREAD

SCIENTIFIC OR ENGINEERING MECHANISM
- CONCURRENCY → overlapping progress
- PARALLELISM → simultaneous execution

OUTPUT / FAILURE BOUNDARY
- RACE / DEADLOCK → coordination failures
- SOURCE → compiler / interpreter / assembler

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PARALLELISM → simultaneous execution; RACE / DEADLOCK → coordination failures.

05

STAGE 05

NETWORK DEVICE AND LAYER MAP

NETWORK DEVICE AND LAYER MAP · IP + ROUTER · WI-FI + SWITCH · APPLICATION → HTTP

APPLICATION → HTTP

TRANSPORT → TCP ⇒ INTERNET → IP + router ⇒ LINK → Ethernet

CONCEPT / ARCHITECTURE
- APPLICATION → HTTP
- TRANSPORT → TCP

MECHANISM / APPLICATION
- INTERNET → IP + router
- LINK → Ethernet

READINESS / STATUS LIMIT
- Wi-Fi + switch
- PHYSICAL → electrical

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: LINK → Ethernet | Wi-Fi + switch; PHYSICAL → electrical | optical | radio.

06

STAGE 06

INTERNET AND WEB REQUEST FLOW

INTERNET AND WEB REQUEST FLOW · DOMAIN --DNS--> IP ADDRESS · BROWSER --HTTP/HTTPS--> WEB SERVICE · BROWSER-HELD STATE · PROTECTED TRANSPORT, NOT TRUTHFUL CONTENT

DOMAIN --DNS--> IP ADDRESS ⇒ BROWSER --HTTP/HTTPS--> WEB SERVICE ⇒ TCP/UDP → transport ⇒ IP → routing ⇒ COOKIE → browser-held state / identifier

STARTING CONDITION / INPUT
- DOMAIN --DNS--> IP ADDRESS
- BROWSER --HTTP/HTTPS--> WEB SERVICE

SCIENTIFIC OR ENGINEERING MECHANISM
- TCP/UDP → transport
- IP → routing

OUTPUT / FAILURE BOUNDARY
- COOKIE → browser-held state / identifier
- HTTPS → protected transport, not truthful content

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: COOKIE → browser-held state / identifier; HTTPS → protected transport, not truthful content.

07

STAGE 07

PERFORMANCE AND WIRELESS MATRIX

PERFORMANCE AND WIRELESS MATRIX · ACHIEVED RATE · DELAY VARIATION · PACKET CELLULAR · VOICE OVER LTE · DISTINCT ROLES · WIRELESS LABEL · NO UNIVERSAL RANGE OR WALL PENETRATION

BANDWIDTH → capacity	THROUGHPUT → achieved rate
LATENCY → delay	JITTER → delay variation
LTE → packet cellular	VoLTE → voice over LTE

SYSTEM / DEVICE / INSTITUTION
- BANDWIDTH → capacity
- THROUGHPUT → achieved rate
- LATENCY → delay

DECISIVE TECHNICAL DISTINCTION
- JITTER → delay variation
- LTE → packet cellular
- VoLTE → voice over LTE

STATUS / UNIT / EVIDENCE LIMIT
- VLC → distinct roles
- WIRELESS LABEL → no universal range or wall penetration

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: WIRELESS LABEL → no universal range or wall penetration.

08

STAGE 08

VIRTUALISATION AND CLOUD BOUNDARY

VIRTUALISATION AND CLOUD BOUNDARY · VIRTUAL HARDWARE + GUEST OS · SHARED HOST KERNEL · MANAGES MANY SERVICES · FIVE NIST CHARACTERISTICS · RESPONSIBILITY SHIFTS

VM → virtual hardware + guest OS

CONTAINER → shared host kernel ⇒ ORCHESTRATION → manages many services ⇒ CLOUD → five NIST characteristics

SYSTEM / DEVICE / INSTITUTION
- VM → virtual hardware + guest OS
- CONTAINER → shared host kernel

DECISIVE TECHNICAL DISTINCTION
- ORCHESTRATION → manages many services
- CLOUD → five NIST characteristics

STATUS / UNIT / EVIDENCE LIMIT
- SaaS → responsibility shifts
- VM → virtual hardware + guest OS

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: IaaS | PaaS | SaaS → responsibility shifts.

09

STAGE 09

DATA AND DISTRIBUTED TRADE-OFF

DATA AND DISTRIBUTED TRADE-OFF · ORGANISED MANAGED DATA · TRANSACTION PROPERTIES · AVAILABILITY/LOCALITY, NOT BACKUP · CAP DURING PARTITION · CONSISTENCY OR AVAILABILITY TRADE · PLACEMENT BY LATENCY AND RESILIENCE · MUST REMEMBER

SYSTEM / DEVICE / INSTITUTION	DECISIVE TECHNICAL DISTINCTION	STATUS / UNIT / EVIDENCE LIMIT
DATABASE / DBMS → organised managed data	REPLICATION → availability/locality, not backup	EDGE / FOG / CLOUD → placement by latency and resilience
ACID → transaction properties	CAP DURING PARTITION → consistency or availability trade	MUST REMEMBER: Computing separates hardware, firmware, operating system, application and...

SYSTEM / DEVICE / INSTITUTION
- DATABASE / DBMS → organised managed data
- ACID → transaction properties

DECISIVE TECHNICAL DISTINCTION
- REPLICATION → availability/locality, not backup
- CAP DURING PARTITION → consistency or availability trade

STATUS / UNIT / EVIDENCE LIMIT
- EDGE / FOG / CLOUD → placement by latency and resilience
- MUST REMEMBER: Computing separates hardware, firmware, operating system, application and...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MUST REMEMBER: Computing separates hardware, firmware, operating system, application and.

10

STAGE 10

LEDGER CLAIM FIREWALL

LEDGER CLAIM FIREWALL · DISTRIBUTED LEDGER · REPLICATED RECORD · LINKED BLOCKS + CONSENSUS · ACCESS MODELS · ORACLE INPUT · CAN BE FALSE · CONDITIONAL CLAIMS

DISTRIBUTED LEDGER → replicated record

BLOCKCHAIN → linked blocks + consensus ⇒ CONSORTIUM → access models ⇒ ORACLE INPUT → can be false

SYSTEM / DEVICE / INSTITUTION
- DISTRIBUTED LEDGER → replicated record
- BLOCKCHAIN → linked blocks + consensus

DECISIVE TECHNICAL DISTINCTION
- CONSORTIUM → access models
- ORACLE INPUT → can be false

STATUS / UNIT / EVIDENCE LIMIT
- NFT / WEB3 / IMMUTABILITY → conditional claims
- CLOSE DISTINCTION: Bit is not byte, RAM is not storage, core count is not performance,...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLOSE DISTINCTION: Bit is not byte, RAM is not storage, core count is not performance,.

11

STAGE 11

ADVANCED COMPUTING AND STATUS RAIL

ADVANCED COMPUTING AND STATUS RAIL · PROCESSOR + ACCELERATOR + INTERCONNECT + SOFTWARE · SOFTWARE CAPABILITY · DEVICE BASE · QUBITS FOR SELECTED PROBLEMS · CONFIDENTIALITY + INTEGRITY + AVAILABILITY · VERIFIED OUTCOME · EVIDENCE LIMIT

HPC → processor + accelerator + interconnect + software ⇒ AI → software capability ⇒ SEMICONDUCTOR → device base ⇒ QUANTUM → qubits for selected problems ⇒ SECURITY → confidentiality + integrity + availability

CONCEPT / ARCHITECTURE
- HPC → processor + accelerator + interconnect + software
- AI → software capability
- SEMICONDUCTOR → device base

MECHANISM / APPLICATION
- QUANTUM → qubits for selected problems
- SECURITY → confidentiality + integrity + availability
- STANDARD → DEMO → DEPLOYMENT → VERIFIED OUTCOME

READINESS / STATUS LIMIT
- EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: State layer, component, protocol,...

MECHANISM / VERDICT — EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: State layer, component, protocol,...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: State layer, component, protocol.

E

EXTRA

SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT · MULTIPLE TASKS MAKE PROGRESS DURING OVERLAPPING TIME. · TASKS EXECUTE SIMULTANEOUSLY. · RACE CONDITION · RESULT DEPENDS ON UNCONTROLLED EXECUTION ORDER. · TASKS WAIT INDEFINITELY FOR RESOURCES HELD BY ONE ANOTHER. · VIRTUAL MEMORY · MAPS PROCESS ADDRESSES TO PHYSICAL MEMORY/STORAGE, ENABLING ISOLATION AND

OPTIONAL SCIENTIFIC / ENGINEERING DEPTH
- Concurrency: multiple tasks make progress during overlapping time.
- Parallelism: tasks execute simultaneously.
- Race condition: result depends on uncontrolled execution order.
- Deadlock: tasks wait indefinitely for resources held by one another.

READINESS, UNIT OR STATUS LIMIT
- Virtual memory: maps process addresses to physical memory/storage, enabling isolation and a
- Routing: selecting paths between networks.
- NAT: translates address information, commonly allowing many private devices to share a public
- DHCP: automatically supplies network configuration.

COMPARATIVE TECHNOLOGY / GOVERNANCE NUANCE
- Content Delivery Network: distributes cached content closer to users.
- Load balancer: spreads requests across servers
- Zero trust: does not grant implicit trust merely because a user/device is inside a network;
- Cloud centralises compute and economies of scale.

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.

END OF MASTER CANVAS • poster embeds this exact master • tiled pages are overlapping crops of the same pixels